Didgeridoo Synthesis Using Timbre Morphing

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Abstract

*This research describes an additive synthesis design for the **didgeridoo** which captures many of its subtle timbral and expressive characteristics. The design uses timbre morphing to model the dynamic spectra of the instrument in Csound. The design is very flexible, and allows vocal sounds, harmonic sweeps and flutter tonguing. Finally, a composed musical excerpt illustrates the expressive features of the design.*



Figure 1. Ash Dargan Playing the Didgeridoo

Introduction

We modelled the *didgeridoo* (see Figure 1) using additive synthesis in Csound (Vercoe 1992), which captures many of its subtle timbral and expressive characteristics. This paper first gives some background for the *didgeridoo* and then describes the source material we used in this research. Following a brief summary of some additive synthesis customizations we have used to model other instruments, we discuss the features in a new design using timbre morphing to model the dynamic spectra of the *didgeridoo*. We then examine musical expression on the *didgeridoo*, including vocal sounds, harmonic sweeps and flutter tonguing. Finally, we conclude with an excerpt from a composition illustrating the expressiveness of the design.

Background

Previous work on modelling instruments has primarily focused on simulating western instruments, especially wind instruments (Risset 1969, Risset and Matthews 1969, Morrill 1977, Horner and Beauchamp 1996, Horner and Ayers 2002). Rodet and Lefèvre (1997) connected the frequencies of two notes with a line segment and morphed the transition using parameter interpolation to give a smooth slur. We previously designed wavetable synthesis models (*i.e.*, group additive synthesis models) of many wind instrument tones that are easy to use and sound like the original individual tones (Horner, Ayers and Law 1999, Horner and Ayers 1998a) and synthesized timbre tremolos and flutter tonguing on wind instruments (Ayers 2004).

Researchers have also studied Asian instruments, including the Chinese *dizi* (Tsai 2003, 2004), the Asian free reed instruments (Cottingham 2004) and large Chinese bells (Braun 2003). We modelled the Chinese wind instruments using a wavetable model similar to the one we used for western wind instruments, and slurred varying numbers of notes in trills for the Chinese *dizi* and Chinese folk wind instruments (Ayers 2003, Ayers and Horner 2004a and b). We have synthesized the Chinese flutes and the *suona* using additive synthesis in Csound (Ayers 2005, Ayers and Horner, 2006).

Researchers have studied the acoustical properties of the didgeridoo (Fletcher 1983, Wiggins 1988) and morphed the sound of a double bass and a *didgeridoo* (Kuchera-Morin 2002).

The *Didgeridoo*

The *didgeridoo* is an Australian buzzed lip instrument traditionally made from the trunk of a stringy bark or eucalyptus tree that has been hollowed out by termites (Hudson 1994). Every tree is uniquely shaped, and the meandering termite pathways in the walls of the instrument help determine the character of its sound. Some instruments have a large bell at the end of the tube where the tree root flares out which adds a rich color to the sound. The instrument is played with the lips buzzing into one open end of the tube (see Figure 2), rather than with the lips in a small mouthpiece, as with the western brass instruments. The lips are much

looser than for the brass instruments, making the sound quite rich and deep. In addition, various mouth sounds affect the timbre. The *didgeridoo* plays a drone tone colored with vocal sounds and harmonics. The *didgeridoo* comes in several sizes which produce different fundamentals for the drones. This paper examines synthesis of representative sounds produced on a *didgeridoo* with a tone pitched at B1 (62.5 Hz).



Figure 2. Didgeridoo Mouthpiece.

The Source Material

We worked from CD recordings by Ash Dargan (Larrikia tribe, Darwin, NT), *Ash, Dust & Dirt* (1999) and *Didgeridoo Made Easy: A Beginner's Guide* (2001). (We obtained his permission to analyze timbres from his CDs to create the designs for this research. We have also corresponded with him regarding the results of this research.) Since the *didgeridoo* doesn't typically play scales, we began the research using *didgeridoo* sounds on just one pitch. Because we wanted to use the resulting instrument in a composition, we started with *Wangurra* (sleep) (Dargan 1999), one of the pieces that used a *didgeridoo* in B1, a key that fit the composition. We then chose a collection of interesting sounds to analyze for the instrument model.

Usable Features of the Previous Model

We found that the additive synthesis model used in our previous work (Ayers and Horner 2006) was a good starting point for this project, but it needed some refinement to produce the expressive playing techniques that are characteristic of the *didgeridoo*. The following summarizes the features that we reused in the new model.

The design uses 63 sine waves to model the rich spectra of the *didgeridoo* timbres (see Figure 3). Each code block produces a sine wave at the required harmonic frequency, with a slight random inharmonicity. Each harmonic has a large variation in its spectral average amplitude which makes the tones sound more lifelike and unique. The attack and decay times become longer as the harmonics get higher, which makes the overall tone get brighter as it gets louder and less bright as it decays. Each harmonic's amplitude envelope is multiplied by a slow random noise and a fast random noise (jitter) and the harmonics are added together. The design avoids aliasing by omitting any harmonics above the Nyquist frequency. We used a time-varying FM vibrato on the frequency of the tone paired with an amplitude modulation vibrato.

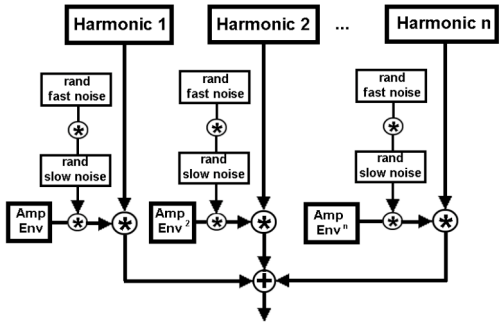


Figure 3. The Additive Synthesis Model.

Synthesizing the *Didgeridoo*

We found the features of the additive synthesis model summarized in the previous section useful in creating a design to model the *didgeridoo*. However, the concept for the new model is different. Since the *didgeridoo* plays basically a single fundamental drone tone and its harmonics, colored by the activities of the voice, we focused on morphing together a string of timbres, rather than slurring a group of pitches.

The following discussion will consider a 2.344 second group of three sounds using the voice to produce "ah-ee-oo too wah," from the beginning of *Wangurra* (Dargan 1999). A spectrogram (Horne 2001) of the segment shows a harmonic sweep at the beginning, followed by a percussive "t" at the beginning of "too" and then a "wah" (see Figure 4).

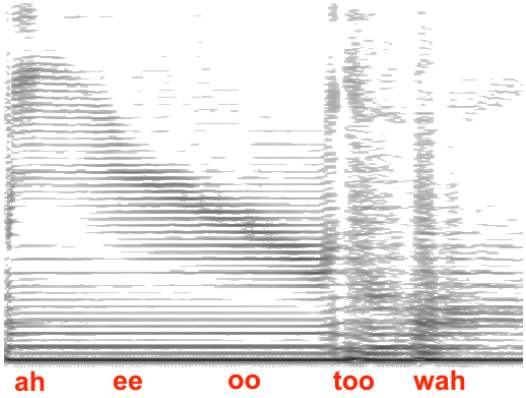


Figure 4. Spectrogram of Original Didgeridoo B1 "ah-ee-oo too wah" Sound Segment.

Figure 5 shows a 63-harmonic time-varying phase vocoder analysis (Dolson 1986, Wun 2000) of the segment. The segment contains a medium amount of instability. The harmonics sweep from the high harmonics at the lower left to the lower harmonics at about time 1.5, when the noisy "t" interrupts the sound, followed by the bright spectrum of the "w" in the "wah."

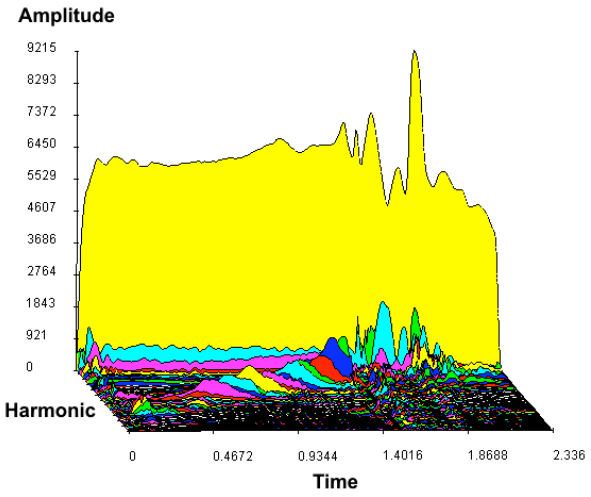


Figure 5. Time-Varying Phase Vocoder Analysis of Original Didgeridoo B1 "ah-ee-oo too wah" Sound Segment.

Figure 6 shows the average spectrum, which has a strong fundamental and a small formant centered on the third harmonic. There is a smaller formant around the 15th harmonic. We found, however, that because the infinite variety of vocal sounds causes the formant to vary considerably, we cannot use the average spectrum to model the *didgeridoo* as effectively as we modelled other wind instruments in our previous additive synthesis designs.

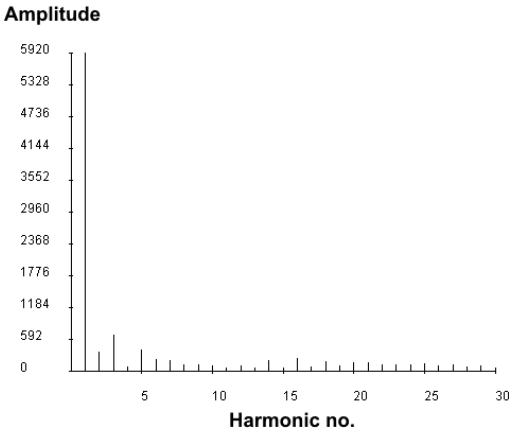


Figure 6. Average Spectrum of Original Didgeridoo B1 "ah-ee-oo too wah"Sound Segment.

For our instrument design, we divided this sound segment into three sounds, a harmonic sweep (the changing vowel sound) and the syllables "too" and "wah." We describe the two syllables in the following sections, as they provide good illustrations for how we modelled the *didgeridoo*. We modelled the harmonic sweep as an expressive technique and we describe it in the section on *Didgeridoo Musical Expression*.

Modelling the "Too" Sound

Figure 7 shows the time-varying phase vocoder analysis of the vocal "too" syllable mixed with the *didgeridoo* tone. This sound has two phonemes, the noisy "t" in the attack and the "oo" vowel which is most of the sound. As our modelling technique involves capturing the spectra of the phonemes and then morphing between them, we might like to call this sound a "morpheme," but as morphemes are sounds with "meaning" and these sounds may be without independent meaning, we will refer to these sounds as "syllables."

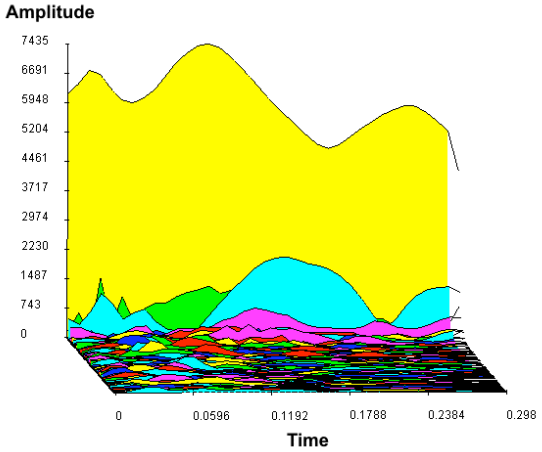


Figure 7. Time-Varying Phase Vocoder Analysis of Original Didgeridoo B1 "Too."

We took two snapshots of this sound (see Figure 8), one for the "t" at time .035 and one for the "oo" at time .267. We chose times from the amplitude peaks in the middle of the section of the waveform that represented the phonemes.

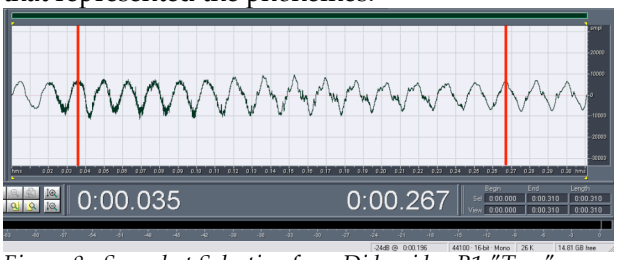


Figure 8. Snapshot Selection from Didgeridoo B1 "Too."

The Spectral Snapshots

After choosing the snapshot times, we examined the phase vocoder analysis graphs (Figures 9-10).

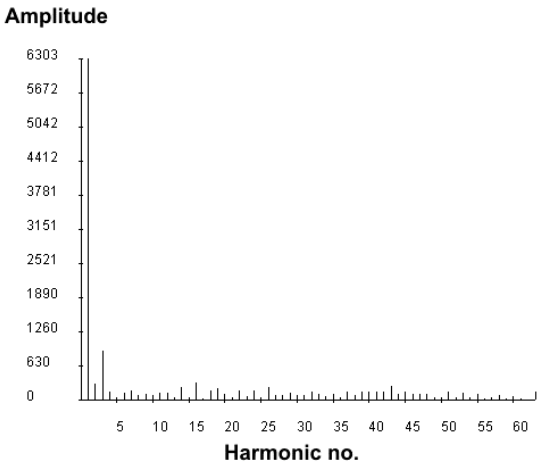


Figure 9. Spectral Snapshot of "t" at Time .035.

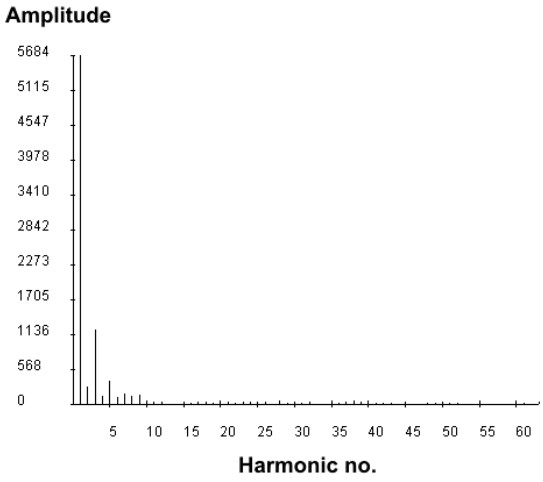


Figure 10. Spectral Snapshot of "oo" at Time .267.

We stored the amplitude data from the graphs in two Csound score function tables. We reserved the number at index 0 in the table for an amplitude at an inharmonic frequency (see the next section, *The Wah Sound).*

```
f51   0    128   -2   h0(v) h1   h2   h3   …
f101  0    128   -2   h0(v) h1   h2   h3   …
```

Although the synthesized "too" sound has a different time-varying spectrum (see Figure 11), the "oo" part does sound reasonably like the original sound. Using this method with one spectral snapshot is unlikely to capture the full quality of the noise in the "t" sound, but the result is sufficient for our musical purposes. An important feature of our design is that repeated performances of the same sound will have varied spectra in order to make the performance expressive.

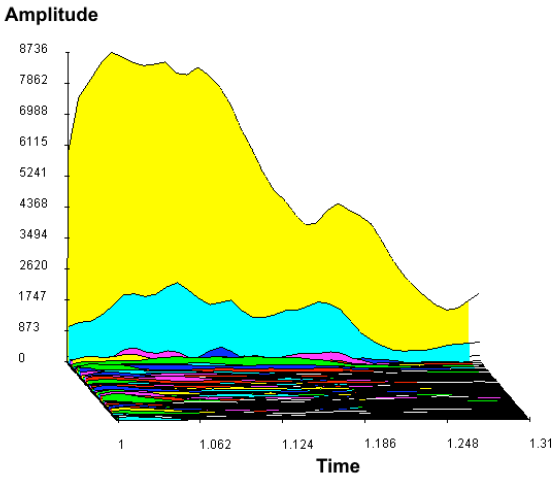


Figure 11. Time-Varying Phase Vocoder Analysis of Synthesized Didgeridoo B1 "too."

The Wah Sound

We divided the "wah" sound into two phonemes, a short "oo" for the "w" and an "ah" (see Figure 12). We chose the snapshot time for the "w" in the unstable beginning at time .049 seconds (see Figure 13). We chose the time for the "ah" near the end at time .659 seconds (see Figure 14) because this part of the sound becomes richer as it begins to morph into the next sound, another harmonic sweep.

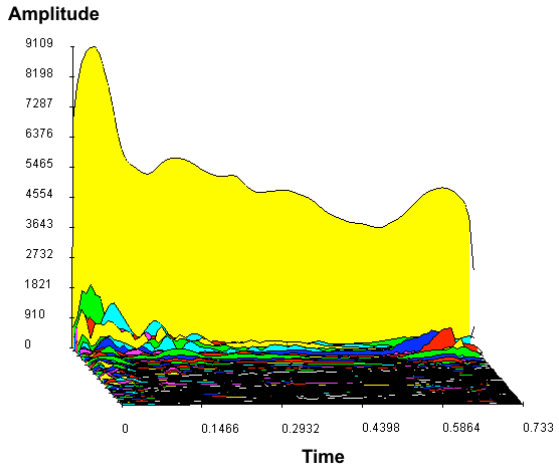


Figure 12. Time-Varying Phase Vocoder Analysis of Original Didgeridoo B1 "wah."

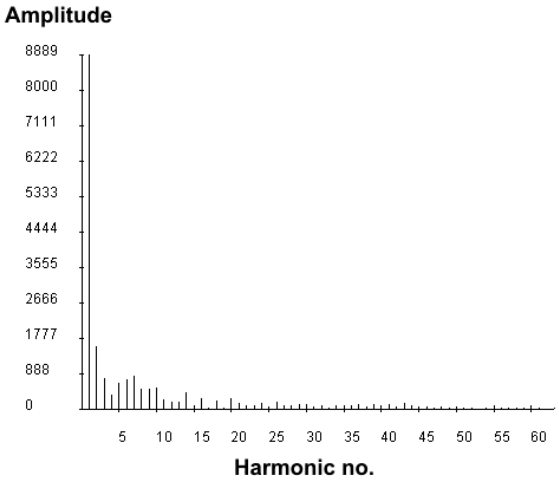


Figure 13. Spectral Snapshot of "w" at Time .049

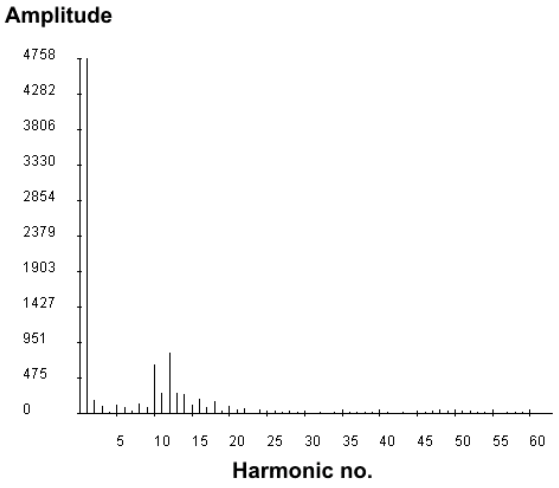


Figure 14. Spectral Snapshot of "ah" at Time .659

The waveform analysis (Hibbert 2003) summary showed that this sound has an extra inharmonic frequency of 174.5 Hz, which is 13 Hz flatter than the third harmonic of 187.5 Hz (see Figure 15).

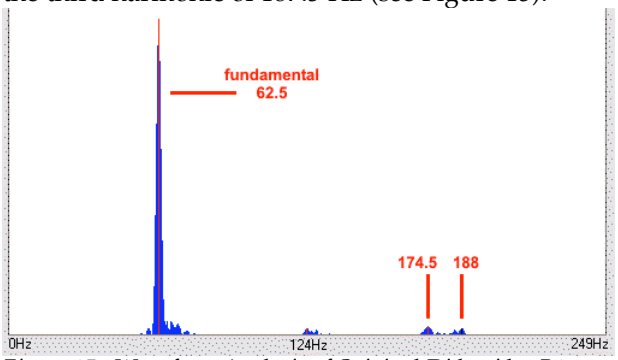


Figure 15. Waveform Analysis of Original Didgeridoo B1 "wah."

Interestingly, however, although the summary doesn't show it, the waveform analysis graph shows the expected harmonic at about 188 Hz and its amplitude is only slightly weaker than the amplitude of the harmonic at 174.5. We added the extra harmonic into the function table at time 0, after calculating its amplitude in relation to the amplitude of the fundamental. We set a slightly varying frequency of about 170-175 for this harmonic to model the natural frequency variations of a performer's voice. Although the synthesized "wah" sound has a different time-varying spectrum (see Figure 16), it sounds more like the original sound than the synthesized "too."

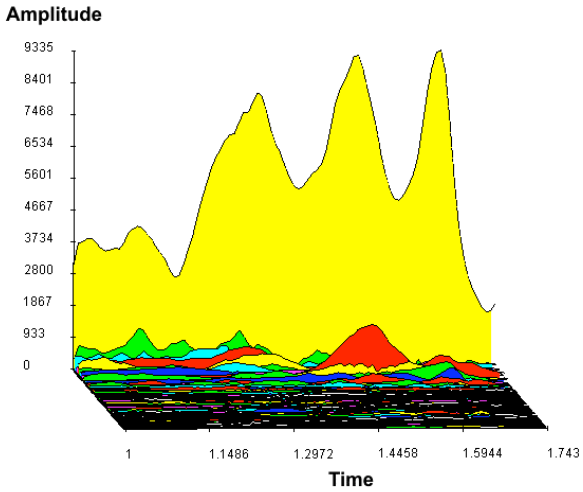


Figure 16. Time-Varying Phase Vocoder Analysis of Resynthesized Didgeridoo B1 "wah."

Musical Expression on the *Didgeridoo*

This section describes vocal sounds, harmonic sweeps and flutter tonguing, expressive techniques that increase the musical potential of the design.

Vocal Sounds

Didgeridoo players use a variety of vocal imitations of animal sounds in their performances. Some of these sounds are the owl, the dingo, the cockatoo parrot, the crow, the buffalo, the kangaroo hop, the kookaburra (laughing jackass), the brolgar (large crane), the emu (Australian ostrich) and the mopoke (boobook owl) (Hudson, 1994, Schellberg 1997, Dargan 2001). The imitation of the buffalo is a particularly gritty sound which uses a combination of vocalization and flutter tongue.

The Harmonic Sweep

The sound segment begins with a vocal sound similar to a smooth owl-like "who" descending from about B4 to E4. The *didgeridoo* resonance pulls the voice toward the *didgeridoo* harmonics similar to the stepped effect a sweeping band-pass filter produces. The resulting sound is more or less similar to the phonemes, "ah," "ee" and "oo." We divided the sound into one pair of slices for each of these phonemes. We chose the snapshot times at .066, .114, .323, .611, 1.136 and 1.487. After we produced a tone by morphing the six sounds, we added a band-pass filter sweeping from the 40th harmonic to the 20th harmonic. This method successfully captures this effect.

Morphing the Sweep Too Wah Sound

We morphed the sounds together to make the combined original sound, "ah-ee-oo too wah." In our previous additive synthesis instruments, we used line segments to connect the note parameters for slurs and morph the transition. In the case of the *didgeridoo*, we don't need to slur "notes," but the code for the slur feature is useful in connecting sound segments of different timbres. Each harmonic is one component signal for the spectrum, with its own frequency and amplitude line segments for the group of morphed notes. Figure 17 shows a Spectrogram analysis of the synthesized "ah ee oo too wah" sound.



Figure 17. Synthesized Harmonic Sweep and "too wah."

Another Harmonic Sweep

Another harmonic sweep moves up and down twice (see Figures 18-19). We found that for musical use of the sweeping band-pass filter effect, it was enough to set the envelopes determining the harmonic sweeping without concern for the exact original vowel sounds. That is, we got a pretty similar effect no matter which original sound we applied it to. Musically, we incorporated the sweeping band-pass filter into the piece with up to 5 (mostly) randomly determined moving segments.

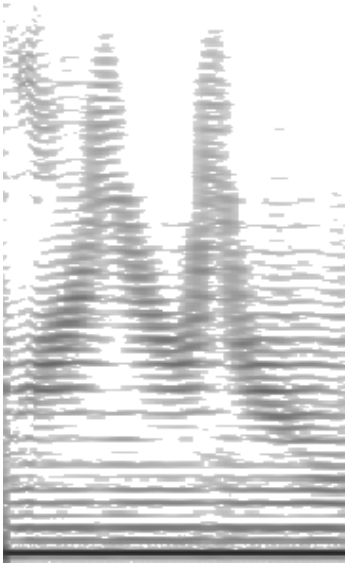


Figure 18. Multiple Harmonic Sweep in Original Didgeridoo B1 Sound.

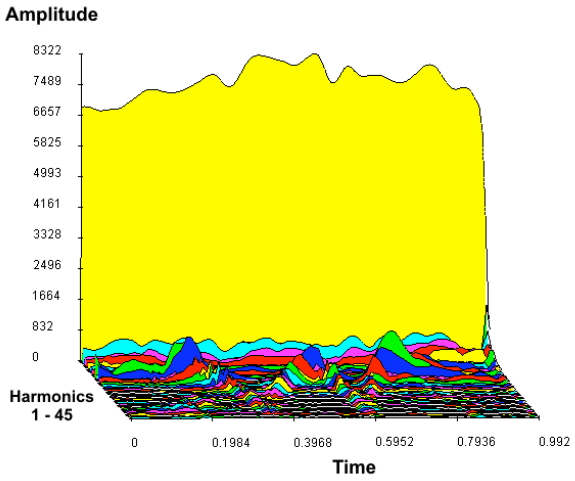


Figure 19. Time-Varying Phase Vocoder Analysis of Multiple Harmonic Sweep on Original Didgeridoo B1 Sound.

Flutter Tonguing

Didgeridoo players typically accomplish flutter tonguing by gargling in the back of the throat, which opens and closes the air stream. The amplitude modulation that we used in the previous wind instrument designs produces a close approximation to this gritty sound.

Examination of Ash Dargan's recorded flutter-tongued *didgeridoo* tones showed that the frequency of the amplitude modulation is approximately 20 Hz. The next step was finding a function to represent the average amplitude envelope of the individual modulations. We found that a simple amplitude envelope would suffice, with a minimum amplitude of 0% of the total amplitude (see Figure 20). The repeating amplitude function models the amplitude change for the flutter. The function does not need to model change of speed or jitter, so it can represent one average cycle of the flutter, and adjusting parameters such as the flutter rate randomly within their typical ranges can vary each cycle. Several score parameters control the flutter rate using a line segment. For example, our design uses four score parameters to control an initial, middle and final flutter rate, and the time required to change from the first flutter rate to the second.

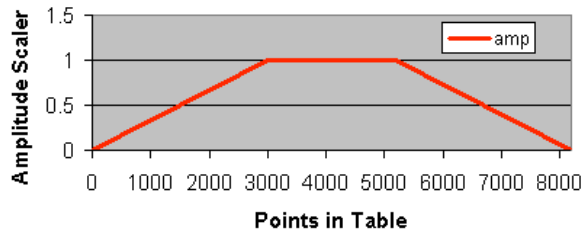


Figure 20. Flutter Tongue Amplitude Envelope Function.

Figures 21 and 22 compare the time-varying phase vocoder analyses of an original B1 *didgeridoo* flutter-tongued tone and a synthesized tone. Although the synthesized tone is less rich than the original, the perceived effect is still quite realistic.

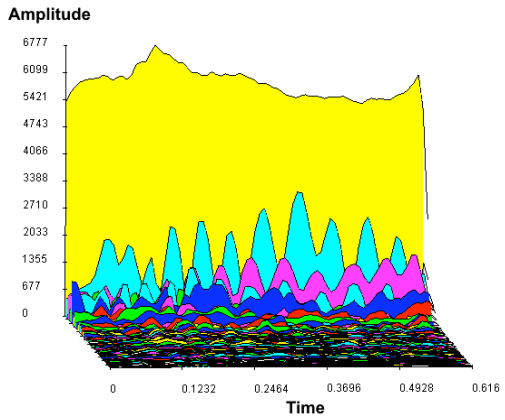


Figure 21. Time-Varying Phase Vocoder Analysis of Original Didgeridoo B1 Flutter-Tongued Tone.

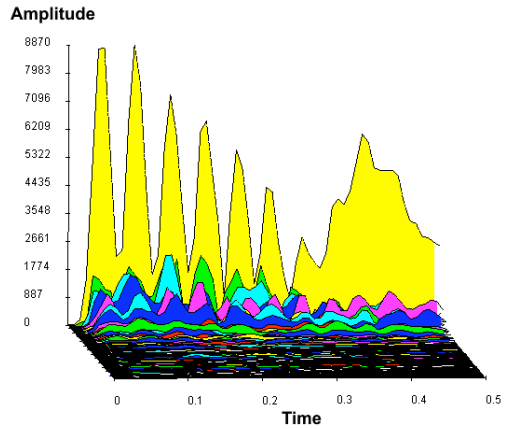


Figure 22. Time-Varying Phase Vocoder Analysis of Synthesized Didgeridoo B1 Flutter-Tongued Tone.

We can either apply the amplitude modulation to the correct part of the sound segment using a delay, or apply it to the entire segment. The instrument easily combines a flutter-tongued timbre with a group of timbres. We use a delay on the amplitude modulator to cause the modulation to begin at the same time as the sound requiring the flutter-tongue occurs in the group of sounds. Using the delay instead of just an amplitude envelope ensures that the flutter begins on the desired timbre.

Use in a Composition

We modelled the *didgeridoo* so we could use it in a composition. Figure 23 shows excerpted *didgeridoo* phrases from the recorded soundscape accompanying a live flute in Ayers' piece, *The Chalky Desert Where Nothing Grows*.

The design currently allows morphing up to 12 pairs of timbres in a connected group. The slurs in the figure indicate typical rhythmic groupings of timbres. As previously described, the flutter tongue may be applied to a single timbre within a group or to a whole group of timbres. We used random cells in a spreadsheet program to choose the *didgeridoo* sounds and vary their parameters.

We then copy and paste chunks of Csound score to increase the input speed. Repeated performances of the same sound will have varied spectra, making the performance more expressive.



Figure 23. Excerpted didgeridoo phrases from Ayers' piece, ***The Chalky Desert Where Nothing Grows***

Conclusion and Future Work

We used additive synthesis with timbre morphing to model the dynamic spectra and musical expression of the *didgeridoo*. The design captures the subtle characteristics of many of its timbres, including vocal sounds, harmonic sweeps and flutter tonguing. Finally, an excerpt from a composition illustrated the expressiveness of the Csound design. Future work includes extending the model to synthesize more vocal sounds combined with *didgeridoo* sounds, along with modelling *didgeridoos* in other keys.

Acknowledgements

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